

Natural Disasters

A natural disaster is the damage or destruction caused by natural forces. It results in a great loss of life and property.

EARTHQUAKE

Earthquake is a natural disaster. It can occur without any warning. The sudden movement under the surface of the earth causes tremors or vibrations on the surface of the earth. It may lead to a minor or major disaster. The place from where an earthquake originates is called the focus. The point directly above the focus on the surface of the earth is called the epicentre. The effect of an earthquake is greatest at its epicentre.



Destruction after an earthquake

Causes

- Most of the earthquakes are caused by the rapid release of energy that is stored within the rocks of the crust.
- Earthquakes occur due to internal pressures within the earth.

Consequences

- It causes a great loss of life and property.
- An earthquake that takes place under the ocean can sometimes cause tsunamis.

It is very difficult to predict when an earthquake will occur. Seismologists have studied the pattern and have outlined some earthquake prone areas.

Japan and Philippines are the most earthquake prone countries in the world.

Precautions

The following precautions should be taken during an earthquake:

- Go to open areas and stay away from buildings.
- If you cannot go out of the building, stand in the corner to avoid things falling on you.
- Do not use a lift during or immediately after an earthquake. Use the staircase.
- Get out of cars or buses if you sense an earthquake.
- If on a road, stay away from trees, poles, electric wires, etc.
- Do not stand near mirrors, shelves, fans, etc.

Major Earthquake

The Bhuj earthquake of 7.6 magnitude (Richter scale) that shook the Indian province of Gujarat on the morning of January 26, 2001 was the most deadly in India's recorded history.

FLOOD

Flood occur when it rains heavily and the area which is usually dry is filled with water.



Flood

Causes

- Floods also occur when a dam collapses or when a sea rises above its level.
- When heavy rain takes place.
- Sudden melting of a large amount of snow on mountains during the summer season.
- Cyclones and tsunamis in coastal areas.
- Collapse or breach of a dam or embankment built across rivers.

- Strong tides in the sea.
- Blocking of the river channel by landslides.

Consequences

- A flood causes huge damage to life and property.
- It causes many water-borne diseases.
- Clean water gets contaminated, which leads to water scarcity.

Floods are common in India during the monsoon season. Some areas get flooded due to their proximity to rivers. Bihar, West Bengal and Assam experience floods almost every year.

Precautions

- Banks of rivers should be repaired if they develop cracks.
- All electrical equipments should be disconnected if there is a danger of flood.

Major Flood

The death toll in the Assam floods on July 5, 2012 rose to 100 even as the water started to recede in most of the 27 affected districts, except Dhemaji. However, the rising water level of the Jiajhal river submerged several villages in the morning. The surging waters also affected parts of National Highway number 52 in the district.

DROUGHT

Drought is a long period of time when there is very little or no rain. Drought affected areas have very few natural sources of water. Farmers are the worst affected during droughts.



Condition of land during droughts

Causes

- Lack of rainfall.
- Insufficient water supply to meet everybody's needs.
- Lack of snowfall in mountain regions (not in India, but in very cold regions of the world).

Precautions

- Reservoirs should be constructed to store rainwater.
- Farmers should be educated to avoid over-cropping and over-grazing.
- Initiatives should be taken by the government to limit human settlements in drought-prone areas.
- Rainwater harvesting, which involves collecting rainwater, should be encouraged.

Major Drought

- In a country like India, where farmers depend on monsoon rains for agriculture, cultivation is not possible in drought conditions.
- Drought causes scarcity of food.
- It also leads to starvation and death of human beings and animals.

India has many drought-prone states, like Maharashtra, Karnataka, Gujarat, Rajasthan, Andhra Pradesh and Orissa.

CYCLONE

A cyclone is a rotating mass of air with minimum pressure in its centre. During a cyclone, wind blows at a very high speed followed by heavy rainfall.



Cyclone



Destruction after a cyclone

Causes

Cyclones are caused by warm tropical moisture bearing clouds, which develop above oceans or seas. They occur in areas of very low pressure, when air heated by the sun rises rapidly and becomes saturated with moisture. This air then condenses into thunder-clouds. When the cyclone continues its course and winds blow strongly, they may cause heavy damage to life and property.

Consequences

- A cyclone causes heavy floods. Flooding may result in the spread of diseases like cholera and malaria.
- It damages the electricity supply system and tele-communication system.

Precautions

- Watch weather forecasts and warnings.
- Move to a safe place before the cyclone hits.
- Do not go near the sea.
- Keep phone numbers of hospitals, relatives and help agencies handy.

Major Cyclone

The coastal areas of Andhra Pradesh were hit by a cyclone in May 1990. Winds blew at a speed of over 250 km per hour. The cyclone affected nearly 6,000 villages. About 10 lakh houses were destroyed.

TSUNAMI

A tsunami is a series of large waves that are created when the water in an ocean is displaced rapidly.



Japan tsunami



Tsunami waves

Causes

Volcanic eruptions, underwater explosions and underwater earthquakes can generate a tsunami. The effects of a tsunami are devastating.

Consequences

Tsunamis can severely affect coastlines, causing large-scale loss to life and property.

Precautions

- People near the sea or the beach should rush to higher places, if a tsunami warning is heard.
- One should stay away from the ocean till it is considered safe by the authorities.

Major Tsunami

On March 11, 2011, off the Pacific coast of Japan, a 9.0 magnitude earthquake produced a tsunami 33 feet (10 m) high waves along Japan's north eastern coast. The wave caused widespread devastation, with an official count of around 20,000 people confirmed to be killed/missing.

VOLCANO

A volcano is a vent or an opening on the earth's crust through which molten materials (magma) comes out on the surface from its interior. The molten magma which flows out through the vent is called lava. Lava comes out when a volcanic eruption takes place. Gases and ashes are forced out of cracks from the crust of the earth. When the volcanic eruption stops and the lava cools, it forms a mountain. A bowl-shaped hollow is formed at the top of such mountains and is called a crater.



Lava flow



Volcanic eruption

Causes

Magma or molten rock is formed beneath the surface due to various reasons. Molten rock ruptures the ground and pours out. Sometimes, it cools down beneath the ground surface instead of pouring out.

Types of volcanoes

- Extinct or dead volcanoes are those which have not erupted for thousands of years and will probably never erupt. For example, Mt. Kilimanjaro in Africa.
- Dormant or sleeping volcanoes are those that have not erupted for many years but have chances of erupting again. When a vent becomes blocked by hardened lava, it becomes a dormant volcano. For example, Mt. Fujiyama in Japan.
- The third and the most dangerous are the active volcanoes. They often erupt and lead to destruction. For example, Mt. Etna in Italy.

We should never forget that an extinct volcano may become active. For example, Mt. Vesuvius in Italy was an extinct volcano, but it erupted after many years and destroyed Pompeii in 79 AD.

Consequences

Volcanoes can destroy entire cities and change the landscape of a place. They cause vast damage to human life and property.

Precautions

- In many places where volcanoes are active, people should move to other far off places till the volcanoes have stopped erupting.
- Food and water should be stored for such emergencies.

Major Volcano

Barren Island, located in the Andaman Sea, is the only confirmed active volcano in South Asia. It erupted recently on May 2, 2006.

KEYWORDS

Earthquake: the sudden movement under the surface of the earth causes trembles or vibrations on the surface of the earth. It may lead to a minor or major disaster.

Tsunami: huge wave in the sea caused by an earthquake.

Flood: is an overflow of water that submerges land.

Drought: is a long period of time when there is very little or no rain.

Cyclones: a cyclone is a rotating mass of air with minimum pressure in its centre.

SUMMARY

- Natural calamities are sudden changes in nature, which cause great damage and destruction.
- Some of the natural calamities we face in our country are floods, earthquakes, drought, landslides, etc.
- Earthquakes cause the surface to tremble and shake, and lead to mass destruction.
- The three types of volcanoes are extinct, active and dormant.
- Areas near the sea coast are prone to cyclones.